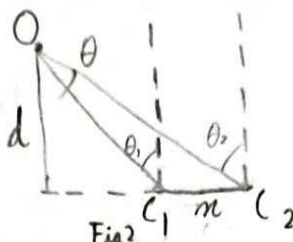


Suppose we have 2 cameras C_1 and C_2 at a distance of π from each other and the object O is at a distance d from the cameras

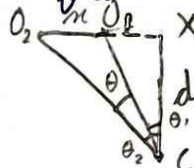
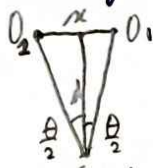


$$\theta + 90^\circ + \theta_1 + 90^\circ - \theta_2 = 180^\circ$$

$$\theta = \theta_2 - \theta_1$$

If we shift the images from the 2 cameras such that it appears as if they are taken

If we overlap the images from the 2 cameras such that it appears as if they are taken from the same point,



$$\theta_2 - \theta_1 = \theta$$

We can see that the 2 ~~objects~~ images appear as if they are at a distance of π and ~~have~~ have an angle of θ with the camera (same as the angle ~~between~~ formed by the 2 cameras with the object)

In fig 4,

Using sine rule in $\triangle CO_1O_2$,

$$\frac{\pi}{\sin \theta} = \frac{CO_2}{\sin(\angle CO_1O_2)} \quad CO_2 = \frac{\pi}{\sin \theta} \times \sin(\angle CO_1O_2)$$

Using sine rule in $\triangle CXO_1$,

$$\frac{CO_2}{\sin(\frac{\pi}{2})} = \frac{d}{\sin(\angle CO_2X)}$$

$$d = \frac{CO_2 \times \sin(\angle CO_2X)}{1}$$

$$= \frac{\pi}{\sin \theta} \sin(\angle CO_1O_2) \times \sin(\angle CO_2X)$$

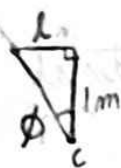
$$= \frac{\pi}{\sin \theta} \sin(90^\circ + \theta_1) \sin(90^\circ - \theta_2)$$

$$= \frac{\pi}{\sin \theta} \cos(\theta_1) \cos(\theta_2)$$

$$d = \frac{\pi \cos \theta_1 \cos \theta_2}{\sin(\theta_2 - \theta_1)}$$

Hence, if we know the angles θ_1 and θ_2 , we can calculate the depth of the object.

To calibrate the camera, hold a stick / ruler of length l (in metres) at a distance of 1m in ~~front~~ front of the camera as shown in the figure and capture an image.



$$\tan \phi = \frac{l}{1}$$

$$\phi = \tan^{-1}(l)$$

Let u be the fraction of width of the image occupied by the stick.

~~$u = \text{No. of pixels occupied by stick horizontally}$~~
Calculate u by,

$$u = \frac{\text{Horizontal length of the stick in pixels}}{\text{Total horizontal length of the image}}$$

The value of u is directly proportional to $\tan \phi$

~~For an object~~

~~For an object at a relative x value~~

For an object at a relative x value (fraction of horizontal length) x_{rel} at an angle θ from the centre of the image.

$$\frac{x_{rel}}{u} = \frac{\tan \theta}{\tan \phi}$$

$$x_{rel} = u \tan \theta$$

$$\tan \theta = \frac{x_{rel}}{u} \tan \phi$$

$$\theta = \tan^{-1}\left(\frac{x_{rel}}{u} \tan \phi\right)$$

$$= \tan^{-1}\left(\frac{x_{rel}}{u} \tan \phi\right)$$

$$= \tan^{-1}\left(x_{rel} \frac{l}{u}\right)$$

$$\text{Take } \frac{l}{u} = k$$

(k is calculated during calibration)

$$\theta = \tan^{-1}(k x_{rel})$$

Let x_n be the relative x value from the left end

$$x_{rel} = x_n - 0.5$$

$$\theta = \tan^{-1}(k(x_n - 0.5))$$



Same u for θ_1 as well as θ_2

Calculate value of k during calibration as $k = \frac{f}{u}$.

Measure the horizontal distance between the cameras as x meters

Train a YOLO model to ~~recognize objects~~ detect objects.

Use the model to detect objects in ~~the~~ both the images in the stereo pair.

The model gives the relative x coordinate values x_r for the centre of the object,

Calculate θ_1 as the θ value for the left camera and θ_2 as that for the right camera.

$$\theta = \tan^{-1}(k(x_r - 0.5))$$

Now we can calculate the depth of the ~~the~~ object d as

$$d = \left| \frac{x \cos \theta_1 \cos \theta_2}{\sin(\theta_2 - \theta_1)} \right|$$

In case the object's bounding box is close to the edges ~~of~~ of the image, ^(in any of the images) we can use the coordinates of the corner points instead to do these calculations.

We choose the corner point farthest away from the edges ~~is~~ for this purpose.

~~For~~ For example, if the centre point of the bounding box is in the top left quadrant of the image, we choose the bottom ^{right} corner of the box.

To get the coordinates of the bottom right corner, we do the following.

$$\begin{aligned} \text{relative } x \text{ coordinate of corner} &= \text{relative } x \text{ of center} + \frac{\text{width of box}}{2} \\ \text{relative } y \text{ coordinate of corner} &= \text{relative } y \text{ of center} - \frac{\text{height of box}}{2} \end{aligned}$$

We use this new value as x_r for the calculation.

Alternatively we can use a machine learning model trained specifically to measure depth from stereo images.

We can concatenate the 2 images vertically.

Then we use a CNN to predict the class of the object, x and y coordinates of the bounding box and also the z value (depth).

We train this on a dataset of stereo images until the prediction is accurate